SuperCLIP-Recon: Masked Caption-Token Reconstruction as an Auxiliary Loss for a COCO-Scale SuperCLIP Baseline

Cameron Caputa
Bocconi University

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Abstract

CLIP-style vision-language models learn transferable representations by aligning images and captions through a global contrastive objective. SuperCLIP strengthens this paradigm by adding classification supervision from raw caption tokens, encouraging the image encoder to preserve more fine-grained semantic information. This project studies whether an additional caption reconstruction objective provides incremental benefit beyond a SuperCLIP-style token-classification baseline. We implement a COCO-scale system using a ViT-B/32 CLIP backbone, a token-classification head, and a lightweight image-conditioned reconstruction head. The baseline is retrieval-grounded: it optimizes both CLIP contrastive loss and token classification, rather than image-only token supervision. The main extension, ReconA, predicts withheld caption tokens from image features. On MS-COCO retrieval, ReconA with $\lambda_{\text{recon}} = 1.0$ changes the five-seed mean R-sum from 445.817 to 446.134, a small average gain of +0.317. However, the paired deltas have high seed variance (+0.317 $\pm$ 0.685 R-sum), are positive for only three of five seeds, and have a 95% confidence interval that crosses zero. ARO and Winoground probes also provide only weak and inconsistent evidence for improved compositional sensitivity. These results suggest that caption reconstruction is feasible and retrieval-compatible, but in this COCO-scale setting it provides limited incremental benefit beyond SuperCLIP-style token classification.

1 Introduction

Contrastive language-image pretraining has become a standard paradigm for vision-language representation learning. CLIP trains image and text encoders by bringing matched image-caption pairs together and separating mismatched pairs in a batch [3]. This global alignment objective supports strong zero-shot and retrieval performance, but it does not explicitly require the image encoder to preserve all fine-grained semantic content expressed in a caption.

SuperCLIP addresses this limitation by adding token-level classification supervision to CLIP [7]. Instead of using only global image-text similarity, SuperCLIP tokenizes captions, converts their tokens into classification targets, and trains an image-side classifier jointly with the CLIP contrastive objective. This provides a simple way to recover supervision from the words in each caption without requiring extra manual labels.

However, token classification remains close to a bag-of-words signal. Captions with similar token sets can describe different actions, attributes, or relations. For example, "a dog chasing a red ball across a field" and "a red ball near a dog sitting in a field" contain overlapping semantic tokens but express different visual situations. This was the central motivation in the project proposal, which argued that token presence supervision misses relational, attributional, and action structure. This limitation is also aligned with prior findings that vision-language models can perform well on retrieval while failing order-, attribution-, or relation-sensitive probes [6, 4].

This project asks whether caption reconstruction can provide useful additional pressure beyond SuperCLIP-style token classification. Rather than replacing contrastive learning or adding a full caption decoder, we attach a lightweight reconstruction head to the image embedding and train it to predict withheld caption tokens. The goal is to test whether reconstruction improves retrieval or compositional sensitivity while preserving the original retrieval-grounded training objective.

Our contributions are: (1) a COCO-scale SuperCLIP-style baseline that keeps CLIP contrastive retrieval training active while adding caption-token classification; (2) an image-conditioned caption reconstruction auxiliary loss, with masked-token reconstruction as the main variant; and (3) an evidence-grounded evaluation on MS-COCO retrieval, limited diagnostic lambda and mask-rate checks, and supporting ARO and Winoground probes. The results show small retrieval-compatible gains but mixed compositional evidence.

2 Related Work

Contrastive vision-language learning. CLIP learns aligned image and text embeddings using a symmetric contrastive objective over image-caption pairs [3]. This objective is effective for zero-shot classification and retrieval, but its supervision is global: each image-caption pair is optimized through a single similarity score. As a result, fine-grained token-level information in captions may be only indirectly learned.

SuperCLIP. SuperCLIP is the closest methodological reference for this project [7]. It augments CLIP with classification supervision from raw caption tokens. Captions are tokenized using the CLIP tokenizer, represented as token-label targets, optionally reweighted using inverse document frequency, and used to train an image-side classification head. The objective is

$$L_{\mathrm{SuperCLIP}} = L_{\mathrm{CLIP}} + L_{\mathrm{Class}}. \tag{1}$$

SuperCLIP reports consistent improvements over CLIP across zero-shot classification and image-text retrieval. Our work does not reproduce SuperCLIP at its original scale. Instead, we use a COCO-scale SuperCLIP-style baseline and test whether reconstruction provides incremental benefit beyond token classification.

Other CLIP extensions address fine-grained or compositional weaknesses through different mechanisms, such as hard-negative caption construction, local region-token alignment, or auxiliary self-supervised objectives [6, 5]. These approaches can provide stronger relational supervision, but they often require additional data construction, architectural changes, or more expensive alignment machinery. Our goal is narrower: we test whether a minimal image-conditioned reconstruction head can add useful pressure beyond token-presence supervision while preserving the SuperCLIP-style retrieval objective.

Compositional vision-language evaluation. ARO and Winoground evaluate whether models distinguish fine-grained compositional alternatives [6, 4]. ARO targets attribution, relation, and order sensitivity, while Winoground uses paired images and captions whose correct matching depends on word order and compositional structure. These benchmarks are relevant because reconstruction is motivated by the gap between token presence and relational semantics.

Masked prediction. Masked language modeling predicts withheld tokens from context and has been influential in representation learning. ReconA is only loosely inspired by this idea: instead of predicting masked text tokens from surrounding text, it predicts selected caption tokens from image features. This makes reconstruction an auxiliary visual grounding signal rather than a language-modeling objective.

3 Methodology

3.1 Baseline: SuperCLIP-style token classification

Let x_i be an image and t_i its paired caption. A CLIP image encoder and text encoder produce normalized embeddings

$$u_i = \frac{f_I(x_i)}{\|f_I(x_i)\|_2}, \quad v_i = \frac{f_T(t_i)}{\|f_T(t_i)\|_2}. \tag{2}$$

The CLIP loss L_{CLIP} is the standard symmetric contrastive loss over image-to-text and text-to-image similarities within a batch.

The SuperCLIP-style baseline adds token classification from the image embedding. Captions are tokenized, mapped to a COCO-derived top-K token vocabulary, and converted into multi-hot token targets y_i^{tok}. A classifier h_{tok} predicts token logits from the image embedding:

$$z_i^{\mathrm{tok}} = h_{\mathrm{tok}}(u_i). \tag{3}$$

The baseline loss is

$$L_{\mathrm{base}} = L_{\mathrm{CLIP}} + L_{\mathrm{tok}}(z_i^{\mathrm{tok}}, y_i^{\mathrm{tok}}). \tag{4}$$

This baseline is important: it is not plain CLIP and not image-only token classification. It preserves contrastive image-text retrieval training while adding token-level supervision. It is therefore best described as a COCO-scale SuperCLIP-style baseline, not a full reproduction of SuperCLIP, because it uses a top-K COCO vocabulary rather than full-vocabulary raw token targets with the original large-scale training setup.

3.2 Caption reconstruction extension

The reconstruction extension adds a head h_{rec} that predicts selected caption tokens from the image embedding. The full objective is

$$L_{\mathrm{total}} = L_{\mathrm{CLIP}} + L_{\mathrm{tok}} + \lambda_{\mathrm{recon}} L_{\mathrm{recon}}. \tag{5}$$

The reconstruction head predicts a fixed number of masked-token slots from the normalized image embedding. For each image feature $u_i \in \mathbb{R}^{512}$, the head creates $M = \lfloor 77\, r_{\mathrm{mask}} \rfloor + 1$ reconstruction slots, where 77 is the CLIP context length and r_{mask} is the mask ratio. In the main setting $r_{\mathrm{mask}} = 0.15$, so $M = 12$. Each slot has a learned 64-dimensional slot embedding, which is concatenated with the image feature and passed through a two-layer MLP with hidden dimension 512 to produce logits over the CLIP vocabulary of size 49,408. The head is non-autoregressive and receives no text context; it predicts masked caption tokens only from image features and slot identity.

For selected target tokens m_i, the reconstruction loss is cross-entropy:

$$L_{\mathrm{recon}} = \mathrm{CE}(h_{\mathrm{rec}}(u_i), m_i). \tag{6}$$

Empty slots are padded with target id 0 and excluded from the reconstruction cross-entropy. Slot k is supervised by the k-th selected token in caption order, so the auxiliary task imposes a simple ordered-slot prediction structure rather than set prediction.

Variant A: masked-token reconstruction. Variant A is the main method. A fraction of caption tokens is selected as reconstruction targets, excluding special tokens. The model predicts these withheld token ids from the image embedding. The main setting uses mask ratio 0.15 and $\lambda_{\mathrm{recon}} = 1.0$.

Variant B: phrase reconstruction. Variant B attempts to reconstruct short phrase-like spans from captions. In the executed implementation, this variant is limited and uses lightweight phrase extraction with fallback masking behavior. Because the result archive contains only narrow Variant B follow-up runs, we treat it as exploratory rather than central.

Table 1: Compact implementation details for the main experiments.

Component	Setting
Backbone	CLIP ViT-B/32, OpenAI initialization
Embedding dim	512
Token-classification vocab	top-1000 COCO-derived CLIP tokens
Token loss	normalized multi-hot token classification with frequency reweighting
Reconstruction vocab	CLIP vocabulary, 49,408 tokens
Reconstruction head	slot dim 64, hidden dim 512, 2-layer MLP
Main mask ratio / slots	0.15 / 12 slots
Loss weights	$\lambda_{\text{clip}} = 1$, $\lambda_{\text{tok}} = 1$, λ_{recon} swept; main 1.0
Optimizer	AdamW, lr 10^{-5}, weight decay 0.01
Training	10 epochs, batch size 128, AMP enabled
Evaluation	COCO validation retrieval over 5,000 images

4 Datasets

MS-COCO Captions. The primary dataset is MS-COCO Captions, built on the COCO image dataset [2] and caption annotations that provide multiple human descriptions per image [1]. Training uses COCO training captions, and retrieval evaluation uses the COCO validation split. Evaluation follows the standard five-caption-per-image retrieval setup. We report Recall@1, Recall@5, and Recall@10 for both image-to-text and text-to-image retrieval. We also report R-sum, the sum of these six recall values.

ARO and Winoground. To probe compositional sensitivity, we evaluate ARO VG-Attribution and VG-Relation, and Winoground [6, 4]. These evaluations are supporting diagnostics rather than the primary benchmark. ARO is used to test attribution and relation preferences, while Winoground tests paired image-caption matching with word-order-sensitive examples.

5 Experiments and Results

5.1 Experimental setup

The central comparison is between the SuperCLIP-style baseline and ReconA. Both use ViT-B/32, batch size 128, 10 training epochs, and COCO retrieval evaluation. The main ReconA setting uses $\lambda_{\text{recon}} = 1.0$ and mask ratio 0.15. We report matched-seed results over seeds 101–105.

5.2 Main retrieval results and seed sensitivity

Table 2 reports the main matched-seed retrieval comparison. ReconA improves the five-seed mean R-sum from 445.817 to 446.134, a gain of +0.317. However, this average is small relative to seed variation. The paired R-sum deltas are +0.096, -0.232, +0.380, -0.128, and +1.468, giving a mean $\pm$ standard deviation of +0.317 $\pm$ 0.685. A 95% confidence interval for

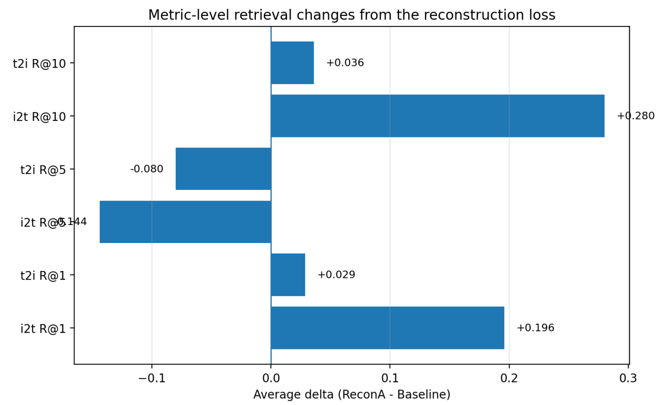

Figure 1: Metric-level retrieval deltas for ReconA − baseline. The average R-sum gain is not uniform across retrieval metrics, supporting the interpretation that ReconA is retrieval-compatible but not consistently better across all metrics.

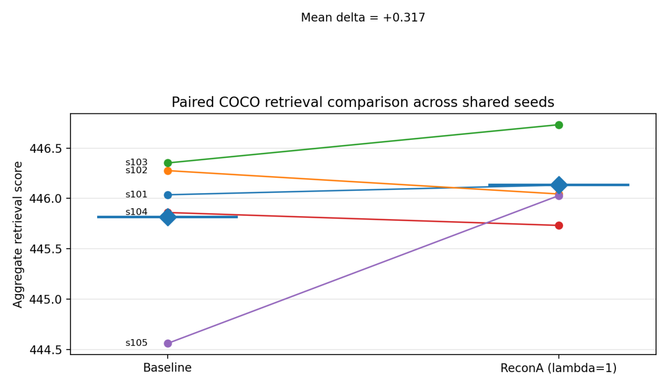

Figure 2: Paired COCO retrieval R-sum for the main matched seeds. The plot shows that the mean gain is small and seed-dependent rather than uniformly positive.

the paired delta is approximately [-0.534, +1.167], so the gain is not statistically established at this scale. Removing seed 105 reduces the mean delta to +0.029, showing that the average improvement is strongly influenced by one low-baseline seed.

Metric-level behavior is also mixed. Image-to-text R@1 and R@10 improve slightly, but image-to-text R@5 and text-to-image R@5 decrease. Figure 1 visualizes these uneven metric-level changes. Thus, the correct interpretation is not that reconstruction robustly improves retrieval, but that it is compatible with retrieval training and may provide a small average gain in this limited five-seed setting.

Figure 2 and Table 3 show paired R-sum changes by seed. ReconA improves three of five seeds and decreases two. The mean gain is partly influenced by seed 105, where the baseline is lower and ReconA improves substantially. This seed-level variability is central to the interpretation of the result.

5.3 Ablations and compositional probes

Table 4 summarizes diagnostic ablations and compositional probes. The lambda comparison is limited to seeds 101–102,

Table 2: Main MS-COCO retrieval results. Values are means over seeds 101–105. R-sum is the sum of image-to-text and text-to-image Recall@1/5/10.

Model	R-sum	I2T R@1	I2T R@5	I2T R@10	T2I R@1	T2I R@5	T2I R@10
SuperCLIP-style baseline	445.817	62.596	86.228	91.840	46.923	74.606	83.623
ReconA, $\lambda = 1.0$	446.134	62.792	86.084	92.120	46.952	74.526	83.659
Delta	+0.317	+0.196	-0.144	+0.280	+0.029	-0.080	+0.036

Table 3: Paired R-sum comparison by seed.

Seed	Baseline	ReconA	Delta
101	446.036	446.132	+0.096
102	446.276	446.044	-0.232
103	446.352	446.732	+0.380
104	445.860	445.732	-0.128
105	444.560	446.028	+1.468

Table 4: Diagnostic ablation and compositional-probe summary.

Experiment	Setting / Metric	Result
Lambda	Baseline, $\lambda = 0$	R-sum 446.156
Lambda	ReconA, $\lambda = 0.5$	R-sum 445.508
Lambda	ReconA, $\lambda = 1.0$	R-sum 446.088
Mask rate	0.15 / 0.30 / 0.50	446.044 / 445.880 / 445.684
ARO	VG-Attribution delta	+0.57
ARO	VG-Relation delta	-0.24
Winoground	Text / Image / Group deltas	+0.25 / +0.67 / +0.50

and the $\lambda = 0$ and $\lambda = 1.0$ rows reuse the corresponding main matched-seed runs rather than independent evidence. In these runs, $\lambda = 0.5$ underperforms the baseline, while $\lambda = 1.0$ is approximately tied with it, making $\lambda = 1.0$ the most reasonable setting to carry forward although the two-seed evidence does not show a clear advantage over the baseline. The mask-rate check is limited to seed 102; in this single-seed diagnostic, higher mask rates did not improve over the main 0.15 setting.

ARO and Winoground provide mixed evidence. ARO attribution improves slightly on average, but ARO relation declines. Winoground scores improve slightly on average, but the gains are small and based on three paired seeds. These results are not sufficient to claim robust compositional improvement.

5.4 Variant B and runtime

Variant B and runtime were treated as follow-ups. On seed 104, the baseline obtains R-sum 445.860; ReconB obtains 446.052 at $\lambda = 0.1$ and 445.832 at $\lambda = 0.5$, which is insufficient to treat Variant B as a main result. Across the five main matched seeds, ReconA increases mean runtime from 3648.3 to 3680.1 seconds, an overhead of 31.8 seconds or about 0.9%. Thus the method is lightweight in observed runtime and architectural disruption, but not parameter-free.

6 Conclusions

This project evaluates caption reconstruction as an auxiliary objective beyond a COCO-scale SuperCLIP-style baseline. The baseline combines CLIP contrastive learning with token classification, making it a retrieval-grounded baseline rather than image-only token supervision. The extension adds masked-token reconstruction from image features.

The main result is modest and statistically uncertain. ReconA with $\lambda_{\text{recon}} = 1.0$ preserves COCO retrieval and gives a small average R-sum increase over five matched seeds, but the paired confidence interval crosses zero and the mean is strongly influenced by one seed. ARO and Winoground results are also mixed, so the experiments do not support a strong claim of improved compositional reasoning. Variant B remains exploratory.

Relative to SuperCLIP, the contribution is incremental: token classification itself is not new, but reconstruction is tested as an additional auxiliary signal beyond token presence. Overall, this project provides a careful negative-to-mixed result: lightweight caption reconstruction can be integrated into a SuperCLIP-style pipeline with little runtime overhead, but stronger claims about retrieval gains or compositional reasoning require larger-scale runs, full-vocabulary SuperCLIP-style targets, stronger phrase reconstruction, and more robust compositional evaluation.

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